

Job Fraud Detection

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1) Problem Objective



Situation:

Job scams are increasingly prevalent in Singapore. Hence, there is a need to **build systems to detect and block suspicious job advertisements** from being published on social media and job platforms.

Objectives:

We have obtained scraped job ad data from social media and want to achieve the following:

1. Identify common trends in fraudulent job postings
2. Develop a model to detect and block suspicious job advertisements





2) Exploratory Analysis



Data Overview

Scraped Job Ad
Data

Categorical
Features

Main Features include:
Industry, Employment Type, Education Level,
Experience Level, Existence of Company Logo

Text Features

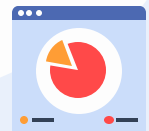
Main Features include:
Company Description, Requirements,
Benefits, Job Title

Numerical
Features

Salary Range

Target Variable

Fraudulent or Non-Fraudulent Job
Posting





2) Exploratory Analysis



Data Quality Assessment



Class Imbalance

Data consists of **17,880 rows**, of which **866 rows** are labelled as **fraudulent**.

Implications:

Risk of high model accuracy but poor fraud detection.

Solution:

Consider methods to balance classes.



High Cardinality Categories

Some features like location (~3000 unique values) have too many different entries.

Implications:

Risk of overfitting model – learning noise instead of useful patterns.

Solution:

Apply feature engineering or drop if it adds little value.



Missing Values

Some features contain missing values.

Implications:

Certain patterns could be hidden due to missing data.

Solution:

Create indicator columns if missingness correlates with fraud or drop columns if it provides little value.



Generalizability

Dataset is not specific to Singapore.

Implications:

Patterns learnt from this data may not fully apply to the local market.

Solution:

Consider collecting or validating on local data before deployment.



2) Exploratory Analysis



What features might be useful for identifying fraud content?

We explore this using a case study for a fraudulent job posting:

Title: Data Entry Office Support **Experience:** Entry Level **Education:** Some High School Coursework

Snippets of Job Description:

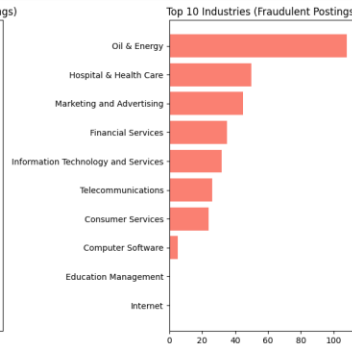
- Hiring employees directly to work from their homes. → Remote opportunity
- Opportunities in a variety of career fields. → Generic job opportunities
- Guarantee wages ranging from \$12 per hour up to six figure salary. → High pay promises with low entry requirements – “Too good to be true”
- This includes bonuses, benefits, and flexibility over your work schedule.
- Training is provided based on the position. → Promised perks and training provided



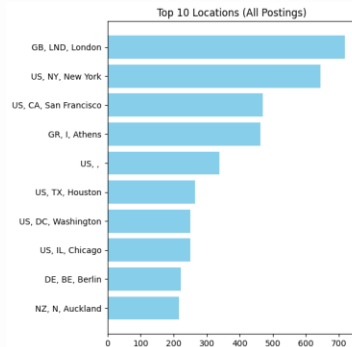
2) Exploratory Analysis



Data trends for Fraudulent VS Non-Fraudulent postings



- Fraudulent postings cluster in **Oil & Energy**, unlike Non-Fraudulent postings which are concentrated in **Information Technology & Services**.
- This contrast suggests **industry type can act as a signal** to help distinguish scams from genuine postings.



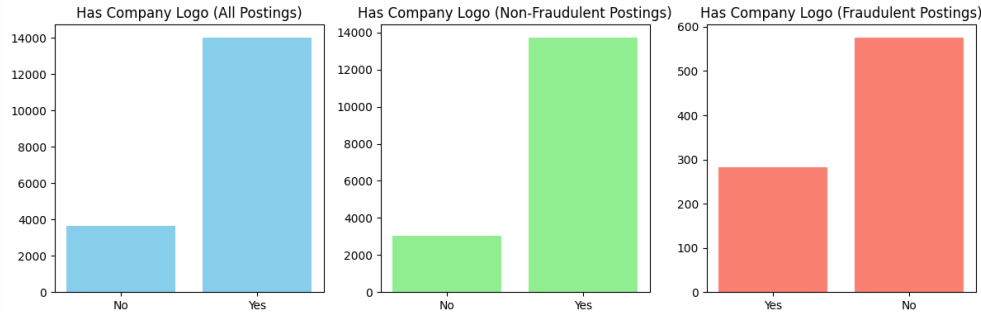
- Fraudulent postings are concentrated specifically in **US, TX, Houston**, while non-fraudulent postings are more widely distributed.
- This suggests that certain **locations may serve as a signal** for detecting scam job ads



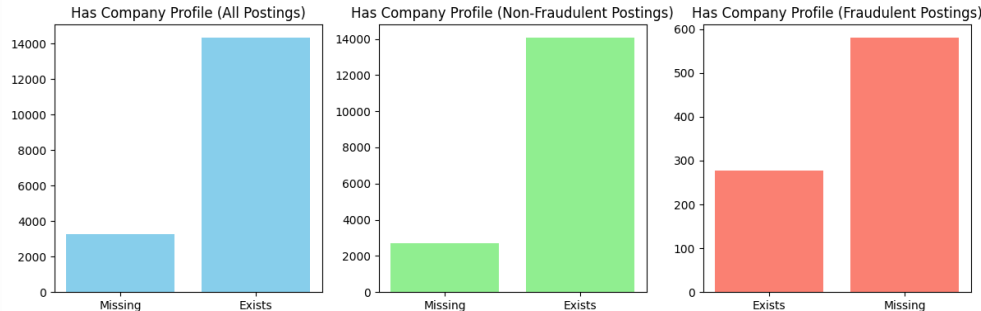
2) Exploratory Analysis



Data trends for Fraudulent VS Non-Fraudulent postings



- Fraudulent postings often **lack a company logo**, while legitimate postings usually include one, making it a **potential sign of legitimacy**.



- Similarly on the existence of a company profile, fraudulent postings often **lack a company profile**, which can be a useful indicator.



2) Exploratory Analysis



Data trends for Fraudulent VS Non-Fraudulent postings

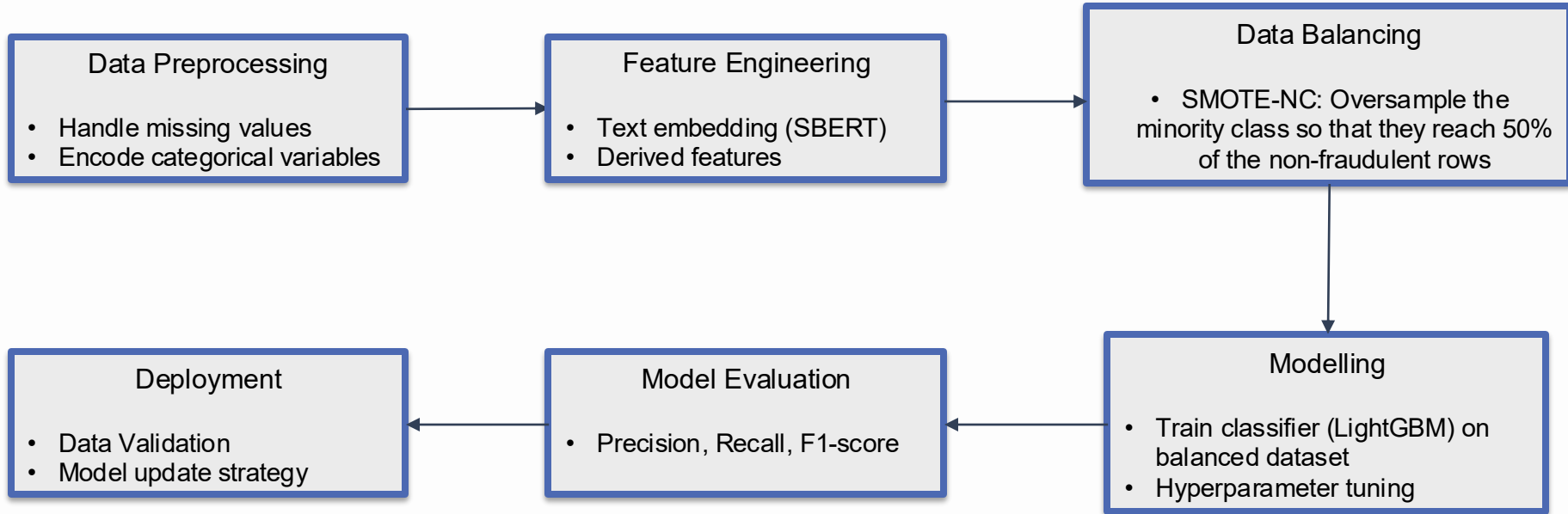
Using TF-IDF text analysis, phrases that were more characteristic of fraudulent job postings compared to non-fraudulent ones were identified:

- Work From Home and urgency cues: *“work home”, “start immediately”*
- Basic skill requirements: *“high school”, “basic computer”, “typing skills”*
- Financial or benefit incentives: *“weekly pay”, “benefits strong”, “online training”*
- Luxury or aspirational language: *“ultra luxury”, “work life balance”, “industry bright future”*

Through exploratory analysis, we see that both categorical and text features are important and informative for fraud detection modelling.



3) Proposed Solution (Overview)





3) Proposed Solution (Main Challenges & Solutions)

Data Processing/ Feature Engineering Step:

- Three main text columns: *description*, *requirements*, *benefits*. Using embedding directly on each column results in very high dimensional data.
- **Solution:** Concatenate the text columns, generate embeddings using SBERT, then perform PCA for dimensionality reduction to simplify the data while preserving information.

Data Balancing Step:

- Dataset is unbalanced – model may favor majority (non-fraudulent) class more.
- **Solution:** Apply SMOTE-NC to increase minority classes. SMOTE-NC is only applied on training data to avoid data leakage.



3) Proposed Solution (Modelling)

Model Recommendation: **LightGBM**

- Handles mixed data types well (Categorical, Text Embeddings)
- Efficient with high dimensional data (Large number of features after text embedding)
- Fast and Scalable: Optimized for large datasets and supports parallel learning
- Provides Feature Importance: Helps to identify which features are most predictive of fraud

Limitation:

- High dimensional data (Text embeddings or many categorical features) can **increase memory usage and training time**. Depending on available resources, we may need to apply more aggressive dimensionality reduction or feature selection.



3) Proposed Solution (Model Evaluation)

Three main proposed metrics:

1. **Recall (Priority):** Measures how many actual fraudulent postings are correctly identified.
2. **Precision:** Measures how many predicted frauds are truly fraudulent, with low precision indicating many false alarms.
3. **F1-score:** Balances Recall and Precision into a single metric.

Goal: High recall, high precision and strong F1-score

Challenge: Trade-off between recall and precision.

Solution: Prioritize maximizing recall (to catch as many fraud cases as possible) and utilize Precision-Recall (PR) curve to select an optimal threshold for classifying fraudulent cases.



3) Proposed Solution (Deployment)

Before deployment:

1. Perform cross-validation to assess model performance using proposed metrics (Recall, Precision, F1-score)
2. Benchmark against prior fraud detection models or published research on job fraud detection
3. Validate model on real-time data before deployment, including manual checks on flagged job postings.

After deployment:

1. Track performance metrics over time (Recall, Precision) to ensure the model remains reliable and effective.
2. Periodically update or retrain model to adapt to emerging fraud patterns and new data trends.